



Companion Cropping and Allelopathic Plants as an Alternative to Chemical Control of Mulberry Pests

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Abstract

Mulberry cultivation faces major threats from insect pests such as mealybugs, thrips, and whiteflies, which limit leaf quality and silkworm productivity. Companion cropping and allelopathic plants offer sustainable, eco-friendly alternatives to chemical pesticides that often cause resistance, environmental pollution, and silkworm toxicity. Companion crops like marigold, basil, and garlic repel pests or enhance natural enemies, while allelopathic plants release bioactive compounds that suppress pest activity. Integrating these practices through intercropping, border planting, and green manuring improves pest management, soil health, and biodiversity. This review highlights their mechanisms, benefits, field applications and future research needs within mulberry-based sericulture systems.

Keywords: Mulberry; Sericulture; Companion cropping; Allelopathy; Allelochemicals; Pest management; Eco-friendly control; Marigold; Garlic; Basil; Neem; Integrated Pest Management (IPM); Biological control; Sustainable agriculture and Mulberry pests.

Introduction

Mulberry (*Morus* spp.) is the exclusive food plant of silkworms, making its health crucial to successful sericulture. However, various insect pests like mealybugs, thrips and whiteflies threaten mulberry cultivation. Chemical pesticides, while commonly used, cause pest resistance, environmental harm, and toxicity to silkworms. These drawbacks have prompted a shift toward eco-friendly pest control strategies. Companion cropping and allelopathic plants offer sustainable alternatives. Companion crops can deter pests or attract beneficial predators. Allelopathic plants release biochemicals that suppress pests without harming the ecosystem. These approaches also improve soil health and biodiversity. As part of integrated pest management (IPM), they help reduce chemical dependency. This article explores their role in mulberry pest control with scientific evidence and practical insights.

Companion Cropping: Concept and Mechanisms

Companion cropping involves the cultivation of two or more plant species in close proximity to benefit at least one of them. This technique can repel pests, attract beneficial insects and improve crop health.

Mechanisms of Pest Suppression

Mechanism	Description
Masking Crop Volatiles	Intercropped species may mask host plant volatiles, confusing pests (Finch & Collier, 2000).
Repellency or Trap Cropping	Some companion crops repel pests or attract them away from the main crop.
Enhancing Natural Enemies	Diverse habitats support predators and parasitoids.
Barrier Cropping	Tall or dense crops create physical barriers against pest movement.

Common Companion Crops in Sericulture

Companion Plant	Mode of Action	Target Pest	Reference
Marigold (<i>Tagetes spp.</i>)	Repellent via terpenes	Whiteflies, aphids	Kumar <i>et al.</i> , 2021
Chrysanthemum (<i>Chrysanthemum cinerariaefolium</i>)	Pyrethrins repel/kill insects	Mealybugs, thrips	Singh & Rao, 2019
Garlic (<i>Allium sativum</i>)	Sulfur volatiles repel pests	Leaf hoppers	Rajasekaran <i>et al.</i> , 2020
Basil (<i>Ocimum spp.</i>)	Volatile oils	Whiteflies	Patel <i>et al.</i> , 2022

Allelopathy

Allelopathy is the biochemical interaction where one plant releases secondary metabolites (allelochemicals) that influence the growth, development, or behavior of other organisms, including insects.

Allelochemicals with Insecticidal Properties

Allelochemical	Source Plant	Effect
Azadirachtin	<i>Azadirachta indica</i> (Neem)	Insect growth regulator
Eugenol	<i>Ocimum sanctum</i> (Holy basil)	Repellent
Pyrethrin	<i>Chrysanthemum spp.</i>	Neurotoxin
Allicin	<i>Allium sativum</i> (Garlic)	Antifeedant

Integration into Mulberry Cultivation

Implementation Strategies

- **Strip Intercropping:** Planting allelopathic species in alternate rows.

- **Border Planting:** Surrounding mulberry fields with insect-repelling crops.
- **Inter-row Planting:** Introducing short-cycle companion crops between mulberry rows.

Examples in Practice

Crop	Intercrop	Impact
Mulberry	Marigold	Reduction in aphids, whiteflies (Nataraja <i>et al.</i> , 2021)
Mulberry	Garlic + Basil	Reduced thrips incidence (Rao <i>et al.</i> , 2020)
Mulberry	Neem-based green manure	Soil pest reduction, growth promotion (Vasudev <i>et al.</i> , 2022)

Comparative Benefits over Chemical Control

Parameter	Chemical Pesticides	Companion/Allelopathic Plants
Pest Resistance	High	Low
Environmental Impact	Negative (pollution, residues)	Positive (eco-friendly)
Safety to Silkworms	Often toxic	Generally safe
Cost	High (recurring expenses)	Low (one-time or seasonal)
Soil Health	Deteriorates over time	Improves organic matter

Challenges and Limitations

- **Knowledge Gap:** Farmers may lack awareness or training on allelopathic systems.
- **Initial Setup:** Selecting compatible species without affecting mulberry yield.
- **Standardization:** Inconsistent results across agro-climatic regions.
- **Labor:** Some companion cropping systems demand more labor input.

Conclusion

Companion cropping and allelopathic plants provide effective, eco-friendly alternatives to chemical pesticides in mulberry fields. They help manage pests naturally by repelling or interrupting pest life cycles. These methods are safe for silkworms and support biodiversity and natural predator populations. Compared to chemical control, they are cost-effective and sustainable for small sericulture farmers. Plants like marigold, garlic, and neem have shown promising results in repelling mulberry pests. Despite their potential, wider adoption requires more research, field trials, and farmer education. Regional compatibility and species selection are key for success. Promoting these practices can align sericulture with organic and climate-smart farming goals. In conclusion, these strategies offer long-term solutions for healthier and more resilient mulberry production systems.

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